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Demystifying Localization

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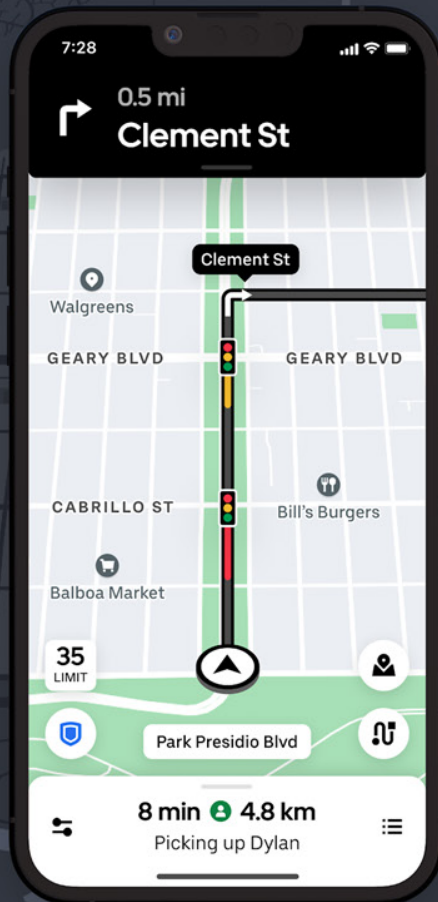
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What is Localization?



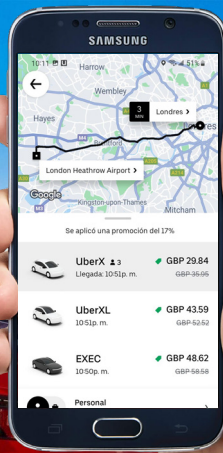
Is that when
the Uber driver has to
find my house?

What is Localization ...really?

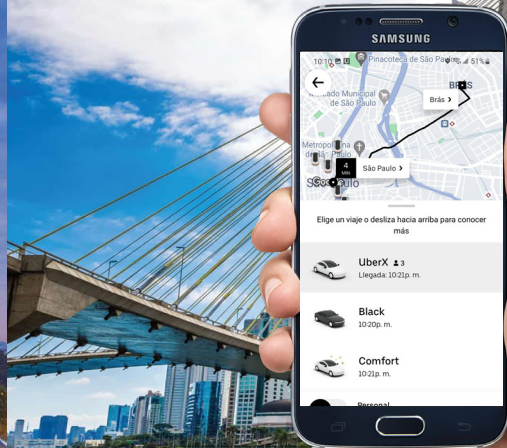
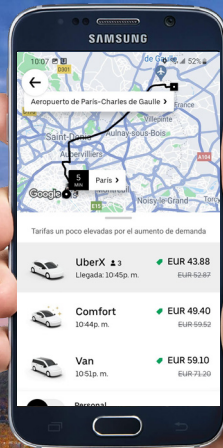
Process of customizing a product for a specific market:

- Language / language variant
- Culture
- Expectations of the local target audience
- Local standards (date format, number delimiters, currency)
- Legal requirements (privacy laws, trademark laws)





Think globally, act locally



Localization at Work

International car launch

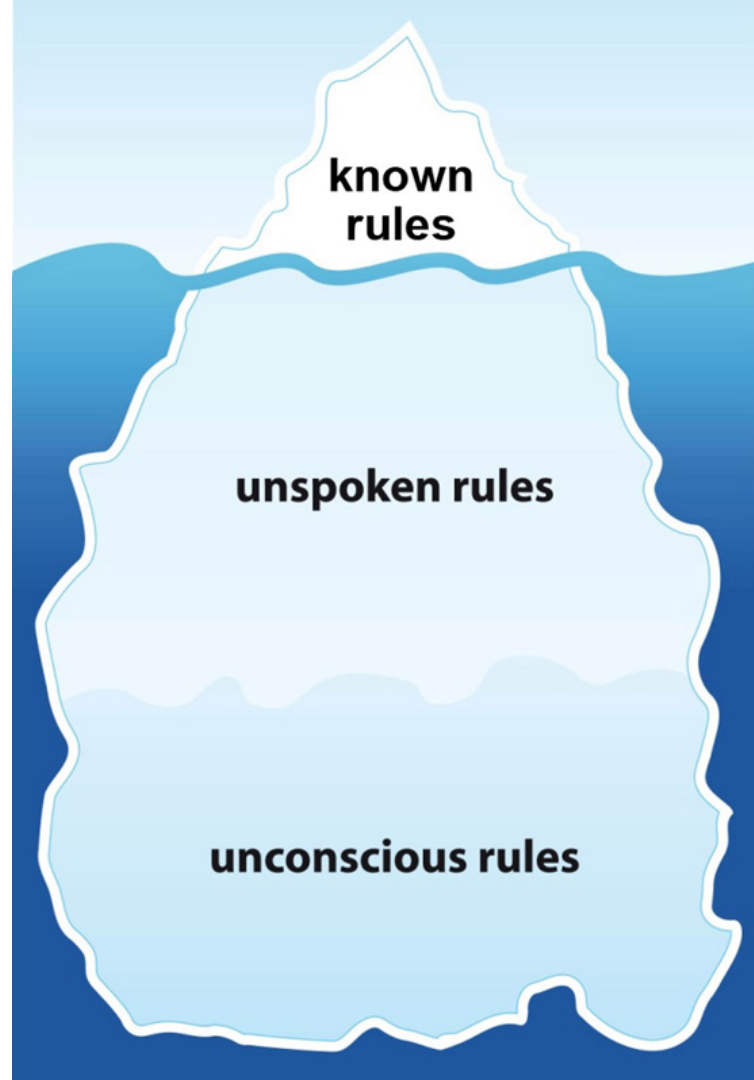
What do we need to think of:

- rename the vehicle model
- offer it in different colors
- localize the infotainment system
- translate the user's manual



Impact of Culture

- **Known rules:**
 - language
 - greetings
- **Unspoken rules:**
 - politeness levels
 - humor
- **Unconscious rules:**
 - understanding language in context



12 ways to Express 'delicious' in Korean

- | | |
|--|---|
| • 맛있네! | 네, 맛있군요 |
| • This is delicious (emphatic) | It really is delicious. |
| • 맛있죠! | 맛있는데요 |
| • This is delicious (seeking validation) | It's surprisingly delicious, contrary to what I had heard |
| • 맛있어! | 맛있어용~~ |
| • This is delicious (informal) | It's delicious [CUTESY] |
| • 맛있습니다 | 맛있당! |
| • This is delicious. (formal) | It's delicious! [CUTESY] |
| • 맛있는데요 | 맛있찌! |
| • Oh, this is delicious. | It's delicious! [CHILDISH] |
| • 맛있대요 | 맛나다! |
| • This is delicious (reported Information, not verified) | It's delicious! [INFORMAL, JOKING] |



GILT: g11n + i18n + l10n + t9n

t9n

Translation

Accurately rendering text from one language into another.

+

l10n

Localization

Adapting a product or content to a specific locale or market.

+

i18n

Internationalization

Designing a product so that it can be adapted to a variety of target markets

g11n

Globalization

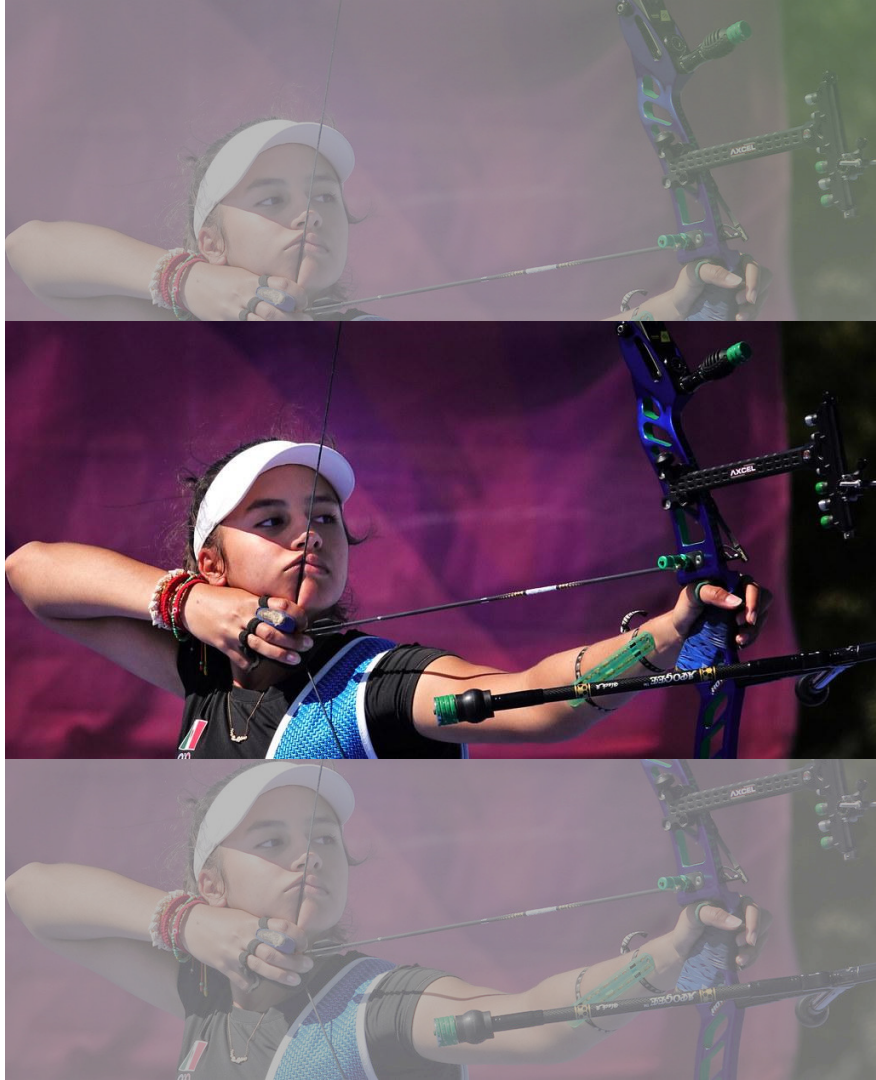
Cumulative process required to develop, market, and scale products internationally.



Source and Target

The original language of the content being localized is called the **source**.

The language(s) the content is being localized into is called the **target**.



Languages vs. Locales

The same “language” has variations depending on “locales”

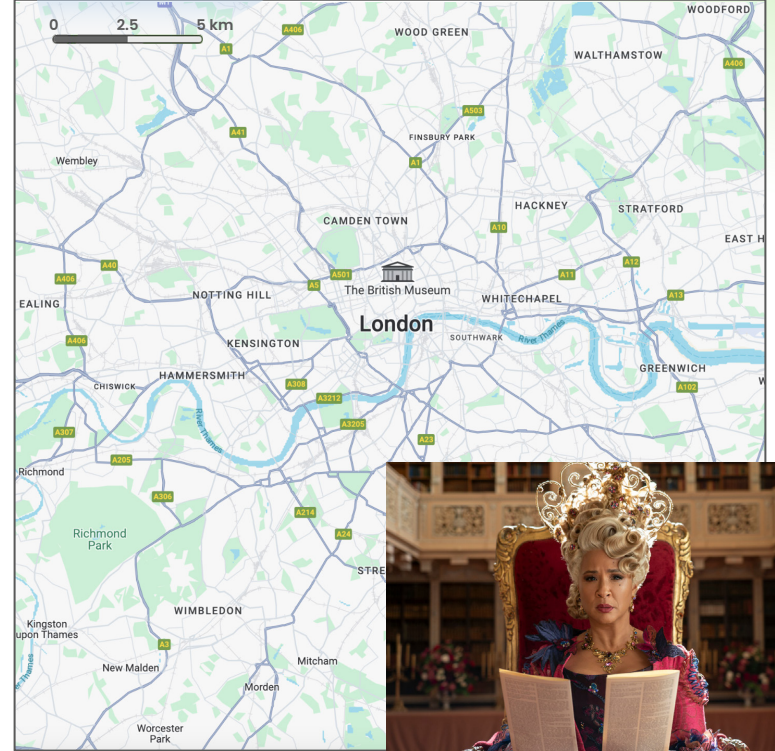
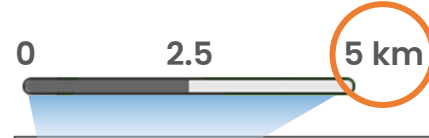
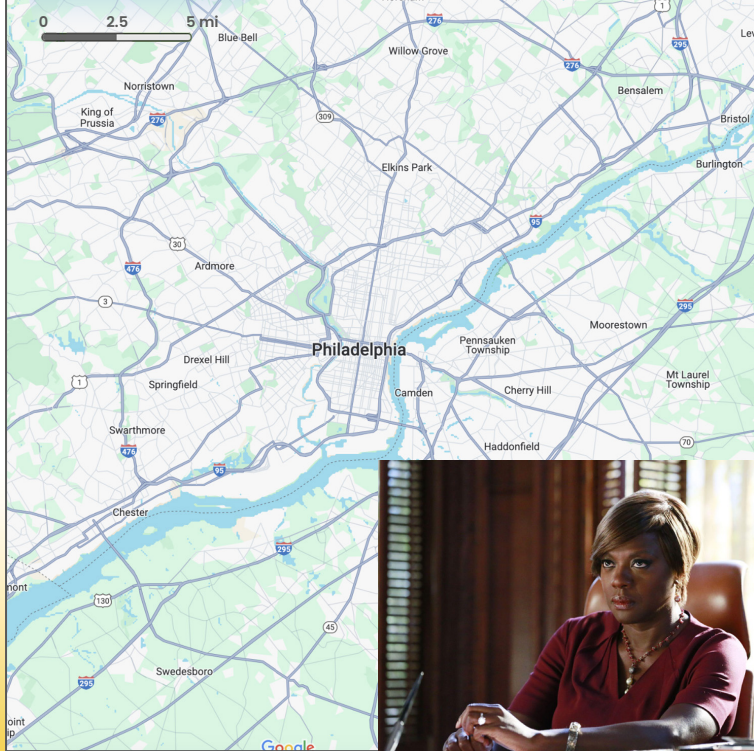
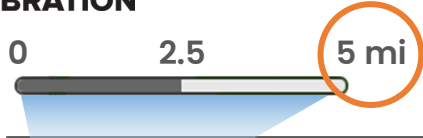
E.g. English has about 160 variations and dialects

Languages & locales are defined in “Language codes”

E.g. en-**US**, en-**GB**, en-**AU**, en-**ZA**, en-**CA**, en-**AE**, en-**TT**, en-**ZW**

Benefits:

- Ensure consistency / interoperability between applications, systems and platforms
- Help prevent target language errors
- Guarantee support of multiple languages and locales



Locale Parameters

Locale specifics are handled by correct i18n + libraries in development

Numbers:

- Different decimal points and thousands separators

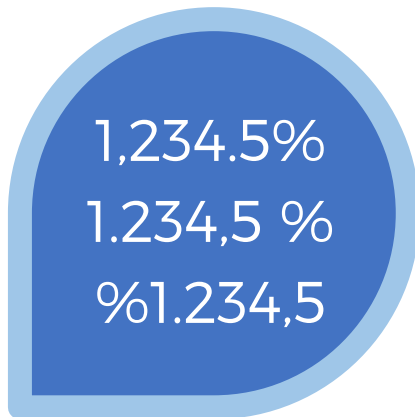
Units of measurement and currency:

- Different abbreviation for units of measurement
- Placement before or after the value they define
- Space between number and unit

Date format:

- Order of month/day/year
- Week starts on Sunday/Monday, weekends vs weekdays

Other locale-based differences: telephone numbers, physical addresses



Localization Tricks

Optimist: The glass is $\frac{1}{2}$ full.

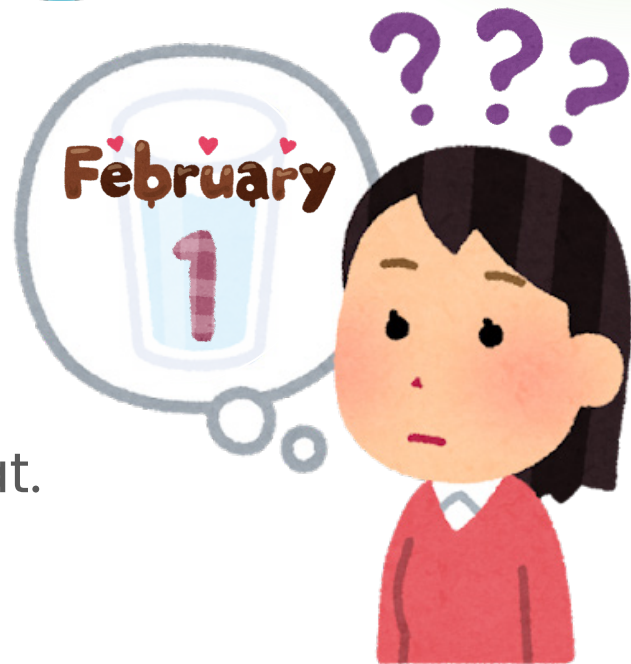
Pessimist: The glass is $\frac{1}{2}$ empty.

Excel: The glass is January 2nd.

Localization: The glass might be February 1st, depending on the date format.



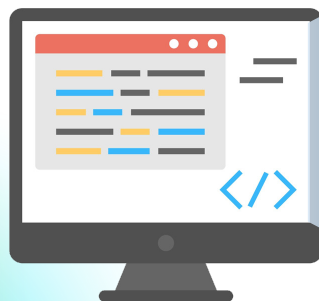
	A
1	1/2
2	2-Jan



Developing with Localizability in Mind

Localizability: readiness of a product for localization

- make strings extractable (avoid hard-coding)
- script support: use Unicode
- avoid concatenations
- use meaningful placeholders & variables



JavaScript Example:

```
let orderDate = "Order Date: 12/31/2024";  
let totalPrice = "Total Price: $1000";
```

Python Example:

```
order_date = "Order Date: 31-12-2024"  
total_price = "Total Price: €1000"
```

Java Example:

```
String orderDate = "Order Date: 2024-12-31";  
String totalPrice = "Total Price: £1000";
```

Avoiding Concatenations

Concatenations: bits of strings joined at runtime in a meaningful unit

Targets have different word order and grammar rules from source



These three strings are not translatable:

"To enable 'Selling" + "in Person"' + "purchase a"

Solution: Provide entire sentences (phrases) with placeholders for variables.

"To enable 'Selling in Person', purchase a {toolkit name}."



Meaningful Placeholder / Variable Names

Placeholders: set of characters representing variables, which are replaced by the actual value

Non-descriptive placeholders are confusing and misleading in L10n:

"{0} will arrive in {1}."

To make them relevant in context, they have to be meaningful:

"{user_name} will arrive in {time_period}."

Define variables in string descriptions, with examples.





Creating Content with Localizability in Mind

International content creation 101:

- conduct global research
- allow for text expansion
- use bias-free, inclusive language
- use culturally neutral icons and colors
- avoid geo-political sensitivities
- use emojis sparingly



**Expand
your Cultural Lens**

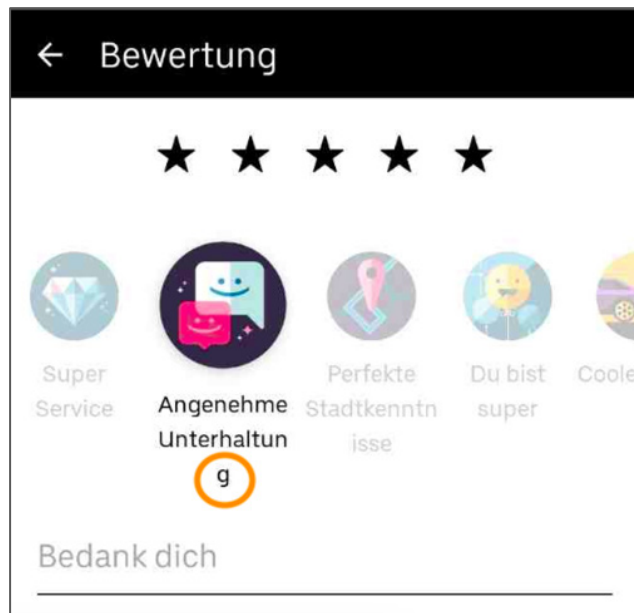


Allow for Text Expansion

Golden rules:

- Leave enough space for the text length to expand
 - by 30% in general
 - by 200%, if the English text < 10 characters
- Define if it should wrap, truncate or scroll
- Stress-test designs with the longest supported language
- Show an early mock to l10n team for feedback

Text Expansion Pitfalls



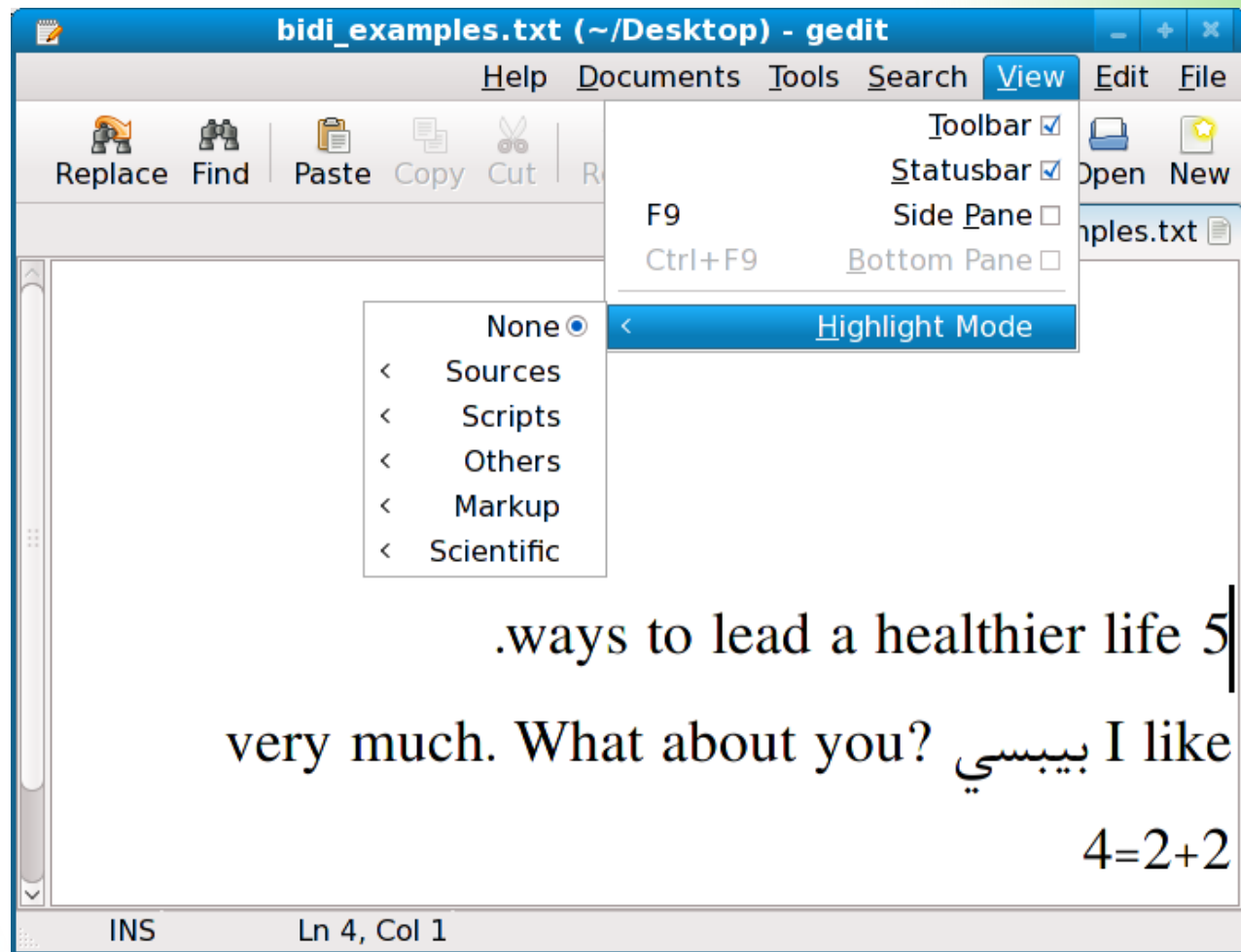
Centering the Global User

Product design is not a salad bar –
we can't just “add a little Localization on it.”

- Involve localization early
- Adapt brand and style guides to target markets
- Use bias-free, inclusive language
- Avoid jargon and slang
- Be careful with humor

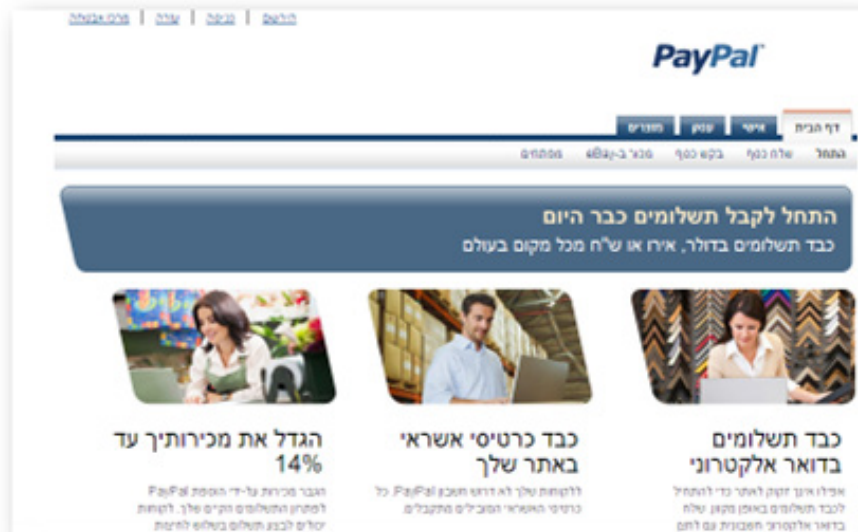


Lack of Planning



Plan International Launches

- Evaluate feature set
- Plan for regional adaptations early
- Share product names with localization



Context is Queen

From reactive to pro-active

In string descriptions

- Pretend that you can't see the UI
- Describe interface elements and interactions
- Explain placeholders and triggers for a string

The diagram illustrates the flow of information from a user interface to a localization system. At the top, a box labeled "Error" represents the UI message. A red arrow points from this box to a specific entry in a localization table. This entry, labeled "1", contains the text "Error" and is linked to a "Context Note" section. The "Context Note" section provides detailed information about the error, including its key, repository, and description. A second red arrow points from the "Context Note" section to a final box at the bottom, which summarizes the repository, key name, and description of the error.

#	Source: en-us	Target: he-il	
JOB # 26	ios-eateronboarding-helix-1.xml		
1	Error	שגיאה	✓ 100
2	Please wait...	יש להמתין...	✓ 100
3	OK	אישור	✓ 100
JOB # 42	web-eats-1.xml		
4	1\${numItems} items, 2\${numGuests} guests	אורחים 2\${numGuests}, פריטים 1\${numItems}	✓
5	From 1\${storeName}	מהמסעדה 1\${storeName}	✓ 100
Context Note			
Key: ff173ac7-a9ef-4bfe-a578-427375488480			
Note: Repo: ios-eateronboarding-helix Key Name: interstitial.error Description: Interstitial Error			

Repo: ios-eateronboarding-helix Key Name: interstitial.error Description: Interstitial Error

Impact of AI on Localization – Benefits

- AI technologies (machine learning, NLP) automate the repetitive translation tasks
- MT and LLM training allows for tailoring translation to specific industries and content types
- Organizations prepare specialized datasets

Key benefits

Efficiency. AI translation engines speed up translation

Cost-effectiveness. Reduced localization costs over time for large-scale projects



Ich habe keinen Bock darauf. ►►► I have no goat on it.



Impact of AI on Localization – Challenges

Content types

Not all content is a good candidate

Specialized fields

Require a level of understanding that AI can't currently match

Translation bias

Likely to perpetuate biases in its training data

Languages

Works less well with languages with non-Latin script

Low-resource languages

Don't have as much data to train MT engines (lower translation quality)

Human oversight required

Quality of AI-driven translations can vary



Partnering with a Language Service Provider (LSP)

When? It was decided that your product is going global

Why? Focus on your core work

Scalability

Consistent quality

How? Work with the existing L10n team

Select a vendor on your own

What are you looking for? A partner that matches your needs

How do you know who's right for you?

- Evaluate their services
- Inquire about their processes
- Make sure the ratio price-quality is right





Formal Education Options

Graduate studies



Middlebury Institute of
International Studies at Monterey

Middlebury Institute of International Studies at Monterey
MA in Translation and Localization Management (TLM)

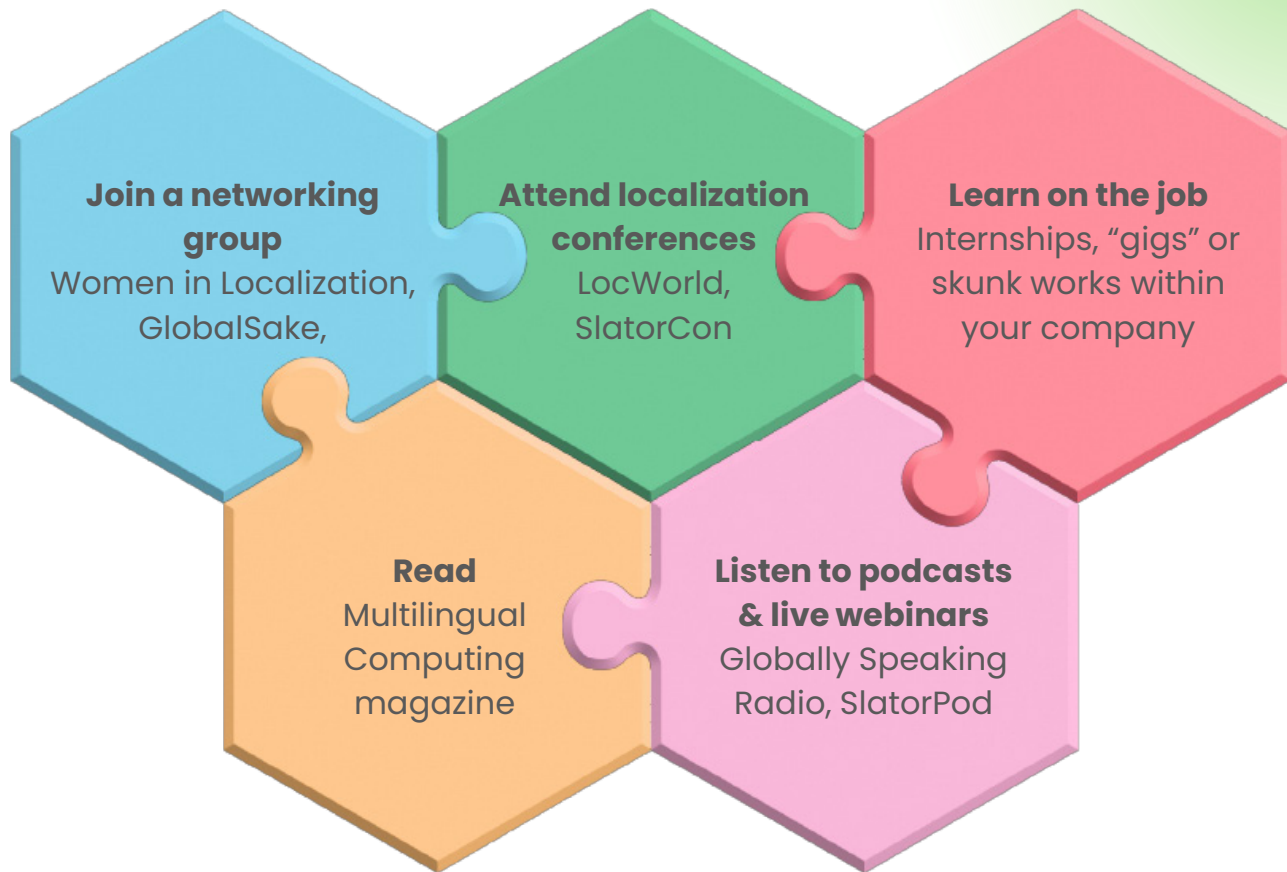


University of Limerick (Ireland)
Localisation Research Centre (LRC)

Other programs

Localization Institute

Informal Education Options



Women in Localization Membership Benefits

Leading professional organization for women in the localization industry

Strives to promote inclusion among all localization professionals

Quarterly
Newsletter

Professional Skill
Development

Localization
Job Board

Invitation to
Events

Mentorship
Program

Localization
Internship
Board

Volunteering
Opportunities

W.L. Partner
Discounts

Access to
Recorded
Webinars



30 Chapters around the world



12,043 members



<https://womeninlocalization.com/>



<https://www.linkedin.com/company/womeninlocalization/>

THANK YOU



<https://www.htpassport.com/>



<https://www.linkedin.com/company/hightech-passport-ltd.>



<https://www.linkedin.com/in/emily-randall-l10n/>



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Questions

